Practical-21

AIM: Perform reading of the data from the MySQL database using PHP in the JSON format and display on the screen of an Android application.

Steps to follow:

1. Start the XAMPP Control Panel
2. Start the apache and mysql server from it
3. In mysql create database user mad@localhost , create database mad , and create table Student with rollno name and sem column
4. Now in xampp/htdocs folder create a php file MAD-Pr20.php and write the code given below
5. Test the php file by typing url in browser with your ip
6. Connect your mobile with same wifi network
7. Test the url in your mobile
8. Finally the app will work

PHP:

```php
<?php

// Set content type to JSON

header('Content-Type: application/json');

// Database connection details

$host = 'localhost'; // or your database host

$db   = 'mad'; // replace with your database name

$user = 'mad'; // replace with your database username

$pass = 'hellomad'; // replace with your database password

// Create connection

$conn = new mysqli($host, $user, $pass, $db);

// Check connection

if ($conn->connect_error) {

   die(json_encode(["error" => "Connection failed: " . $conn->connect_error]));

}

// SQL query to fetch all records from the student table

$sql = "SELECT rollno, name, sem FROM student";

$result = $conn->query($sql);

// Check if records are found

if ($result->num_rows > 0) {

   // Create an array to store the records

   $students = array();

   // Fetch each row and add to the array

   while($row = $result->fetch_assoc()) {

      $students[] = $row;

   }

   // Return the data as JSON

   echo json_encode($students);

} else {

   // Return an empty array if no records found

   echo json_encode([]);

}

// Close connection

$conn->close();

?>
```

XML:

View layout for Recycler View:

```xml
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android
"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="10dp">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/Pr18name"
        android:layout_marginLeft="10dp"
        android:textSize="15sp"
        android:text="Name: Student Name"/>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/Pr18rollno"
        android:layout_marginTop="10dp"
        android:layout_marginLeft="10dp"
        android:layout_below="@+id/Pr18name"
        android:text="RollNo: 226340316099"/>
```

```
   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:id="@+id/Pr18sem"
      android:layout_marginTop="10dp"
      android:layout_marginLeft="10dp"
      android:layout_marginBottom="10dp"
      android:layout_below="@+id/Pr18rollno"
      android:text="Semester: 5"/>
   <View
      android:layout_width="match_parent"
      android:layout_height="1dp"
      android:background="?android:attr/textColorSecondary"
   android:layout_below="@+id/Pr18sem" />
</RelativeLayout>
```

MainActivity:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/Practical21layout"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".Practical21">

   <androidx.recyclerview.widget.RecyclerView
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:id="@+id/Pr21Recyclerview"/>

</RelativeLayout>
```

JAVA:

Student Class (DataHolder):

```
package com.mk.madpractical;

import java.util.Objects;

public class Student {
   String key; // Firebase key
   String rollno;
   String name;
   String sem;

   public Student(String key, String rollno, String name, String
sem) {
      this.key = key;
      this.rollno = rollno;
      this.name = name;
      this.sem = sem;
   }

   public Student() {
      // Default constructor required for calls to
DataSnapshot.getValue(Student.class)
   }

   // Getter and setter for key
   public String getKey() {
      return key;
   }

   public void setKey(String key) {
      this.key = key;
   }

   public String getRollno() {
      return rollno;
   }

   public void setRollno(String rollno) {
      this.rollno = rollno;
   }

   public String getName() {
      return name;
   }

   public void setName(String name) {
      this.name = name;
   }

   public String getSem() {
      return sem;
   }

   public void setSem(String sem) {
      this.sem = sem;
   }
}
```

StudentHolder (ViewHolder for Recycler View):

```
package com.mk.madpractical;
import android.view.View;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

public class StudentHolder extends RecyclerView.ViewHolder {
   TextView name,rollno,sem;
   public StudentHolder(@NonNull View itemView) {
      super(itemView);
      name = itemView.findViewById(R.id.Pr18name);
      rollno = itemView.findViewById(R.id.Pr18rollno);
      sem = itemView.findViewById(R.id.Pr18sem);
   }
}
```

StudentAdapter (Adapter Class for Recycler View):

```
package com.mk.madpractical;
```

```java
import android.content.Context;
import android.view.LayoutInflater;
import android.view.ViewGroup;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

import java.util.List;

public class StudentAdapter extends
RecyclerView.Adapter<StudentHolder> {
   public StudentAdapter(Context context, List<Student>
students) {
      this.context = context;
      this.students = students;
   }

   Context context;
   List<Student> students;
   @NonNull
   @Override
   public StudentHolder onCreateViewHolder(@NonNull
ViewGroup parent, int viewType) {
      return new
StudentHolder(LayoutInflater.from(context).inflate(R.layout.pr
18recyclerview,parent,false));
   }

   @Override
   public void onBindViewHolder(@NonNull StudentHolder
holder, int position) {
      Student student = students.get(position);
      holder.rollno.setText("RollNO: "+student.rollno);
      holder.name.setText("Name: "+student.name);
      holder.sem.setText("Semester: "+student.sem);
   }

   @Override
   public int getItemCount() {
      return students.size();
   }

   public void addStudent(Student student) {
   if (getStudentPositionByKey(student.key) == -1) {
      students.add(student);
      notifyItemInserted(students.size() - 1);
   }
}
   public void updateStudent(Student updatedStudent) {
      int position =
getStudentPositionByKey(updatedStudent.key);
      if (position != -1) {
         students.set(position, updatedStudent);
         notifyItemChanged(position);
      }
   }
   public void removeStudent(String key) {
      int position = getStudentPositionByKey(key);
      if (position != -1) {
         students.remove(position);
         notifyItemRemoved(position);
      }
   }
   private int getStudentPositionByKey(String key) {
      for (int i = 0; i < students.size(); i++) {
         if (students.get(i).getKey().equals(key)) {
            return i;
         }
      }
      return -1;
   }
}
```

Main Activity:

```java
package com.mk.madpractical;

import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;
import android.util.Log;
import android.widget.Toast;
import org.json.JSONArray;
import org.json.JSONException;
import org.json.JSONObject;
import java.io.IOException;
import java.util.ArrayList;
import java.util.List;
import okhttp3.Call;
import okhttp3.Callback;
import okhttp3.OkHttpClient;
import okhttp3.Request;
import okhttp3.Response;
import java.util.ArrayList;

public class Practical21 extends AppCompatActivity {
   RecyclerView recyclerView;
   private ArrayList<Student> students;
   private StudentAdapter adapter;
   private static final String URL_STRING =
"http://192.168.66.127:5700/MAD-Pr21.php"; // Replace with
your PHP URL
   private final OkHttpClient client = new OkHttpClient();
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_practical21);

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R
.id.Practical21layout), (v, insets) -> {
         Insets systemBars =
```

```
insets.getInsets(WindowInsetsCompat.Type.systemBars());
        v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
        return insets;
    });
    recyclerView = findViewById(R.id.Pr21Recyclerview);
    recyclerView.setLayoutManager(new
LinearLayoutManager(this));
    students = new ArrayList<Student>();
    adapter = new StudentAdapter(this, students);
    recyclerView.setAdapter(adapter);
    fetchStudentData();
  }
  private void fetchStudentData() {
    // Build the request to fetch data from the server
    Request request = new Request.Builder()
        .url(URL_STRING)
        .build();
    // Make an asynchronous call using OkHttp
    client.newCall(request).enqueue(new Callback() {
      @Override
      public void onFailure(Call call, IOException e) {
        Toast.makeText(Practical21.this, "Error fetching the
data", Toast.LENGTH_SHORT).show();
      }
      @Override
      public void onResponse(Call call, Response response)
throws IOException {
        if (response.isSuccessful()) {
          final String responseData =
response.body().string();
          runOnUiThread(() -> {
            try {
              parseJsonToStudentList(responseData);
            } catch (JSONException e) {
              Log.e("MainActivity", "Error parsing JSON", e);
            }
          });
        }}
    });
  }

  // Method to parse the JSON response and return a list of
Student objects
  private List<Student> parseJsonToStudentList(String
jsonData) throws JSONException {
    List<Student> studentList = new ArrayList<>();
   // Parse the JSON array from the response
    JSONArray jsonArray = new JSONArray(jsonData);

    // Loop through each object in the JSON array
    for (int i = 0; i < jsonArray.length(); i++) {
      JSONObject studentJson = jsonArray.getJSONObject(i);

      // Extract the values and create a Student object
      String rollno = studentJson.getString("rollno");
      String name = studentJson.getString("name");
      String sem = studentJson.getString("sem");

      // Create the Student object, keeping key same as rollno
      Student student = new Student(rollno, rollno, name,
sem);
      // Add the student object to the list
      adapter.addStudent(student);
      studentList.add(student);
    }

    return studentList;
```

Output:


16:02
1 device
44%
Name: studenta
RollNO: 123789
Semester: 5
Name: Neha Kavaiya
RollNO: 2263403160
Semester: 5
Name: student1
RollNO: 748595
Semester: 5
Name: student2
RollNO: 748596
Semester: 5
Name: student3
RollNO: 748597
Semester: 5
Name: student4
RollNO: 748598
Semester: 5
Name: student5
RollNO: 748599
16:02
1 device
44%
Name: hello
RollNO: 123456789
Semester: 7
Name: hello2
RollNO: 123456789123
Semester: 6
Name: studentb
RollNO: 123781
Semester: 5
Name: studentc
RollNO: 123782
Semester: 5
Name: studentd
RollNO: 123783
Semester: 5
Name: studente
RollNO: 123784
Semester: 5
Name: studentf
RollNO: 123785
Semester: 5

Practical-22

AIM: Integrate Google maps API to your Android application and display your current location in the app.

Steps to follow:

1. Create a new project
2. Then right click on app > new > google > google maps activity
3. In Manifeast file you see a todo and in paste your api key in the api key section
4. You can use below api key if you don't have any
5. <meta-data
   android:name="com.google.android.geo.API_KEY"
   android:value="AIzaSyDV2_xy58r15K6TskZy4KWMuhUDVq67jqM" />

XML:

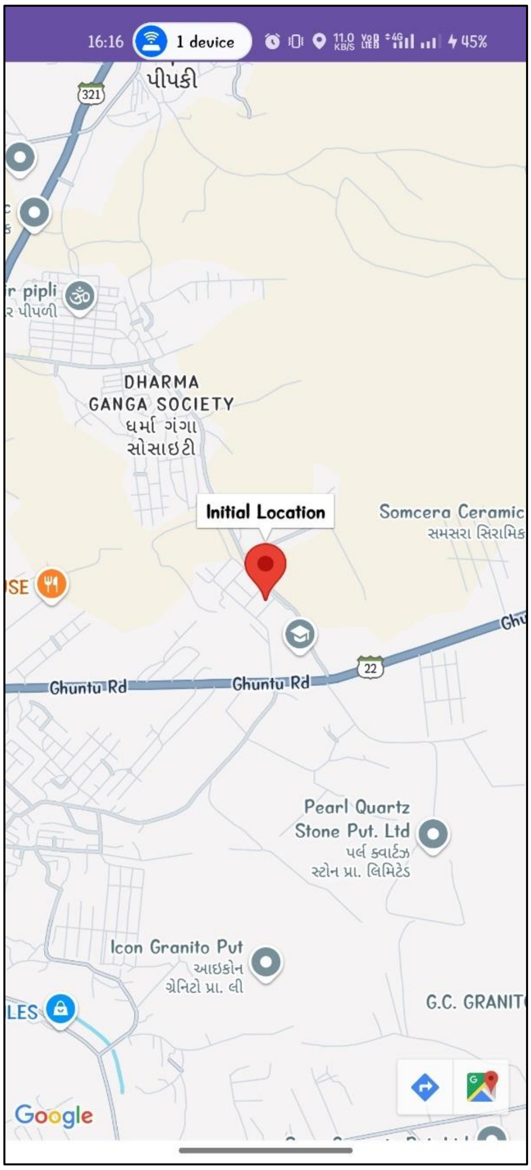
```
<?xml version="1.0" encoding="utf-8"?>
<fragment
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:map="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/map"

android:name="com.google.android.gms.maps.SupportMapFragme
nt"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".Practical22" />
```

JAVA:

```
package com.mk.madpractical;

import androidx.activity.result.ActivityResultLauncher;
import androidx.activity.result.contract.ActivityResultContracts;
import androidx.annotation.NonNull;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
import androidx.fragment.app.FragmentActivity;
import android.content.Context;
import android.content.Intent;
import android.content.pm.PackageManager;
import android.location.Location;
import android.location.LocationListener;
import android.location.LocationManager;
import android.os.Bundle;
import android.widget.Toast;
import com.google.android.gms.maps.model.Marker;
import com.google.android.gms.maps.CameraUpdateFactory;
import com.google.android.gms.maps.GoogleMap;
import com.google.android.gms.maps.OnMapReadyCallback;
import com.google.android.gms.maps.SupportMapFragment;
import com.google.android.gms.maps.model.LatLng;
import com.google.android.gms.maps.model.MarkerOptions;
import com.mk.madpractical.databinding.ActivityPractical22Binding;

public class Practical22 extends FragmentActivity implements
OnMapReadyCallback {
   private static final int LOCATION_PERMISSION_REQUEST_CODE =
1;
   private GoogleMap mMap;
   private LocationManager locationManager;
   private ActivityPractical22Binding binding;
   private ActivityResultLauncher<Intent> locationLoncher;
   private Marker currentMarker;
   private final LocationListener locationListener = new
LocationListener() {
      @Override
      public void onLocationChanged(@NonNull Location location) {

         LatLng newLocation = new
LatLng(location.getLatitude(),location.getLongitude());
         currentMarker.setPosition(newLocation);
         // Optionally, move the camera to the new position

mMap.moveCamera(CameraUpdateFactory.newLatLngZoom(newLo
cation, 15));

      }
      @Override
      public void onStatusChanged(String provider, int status, Bundle
extras) {
      }
      @Override
      public void onProviderEnabled(@NonNull String provider) {
      }
      @Override
      public void onProviderDisabled(@NonNull String provider) {
         Toast.makeText(getApplicationContext(), "Please enable GPS
and Internet", Toast.LENGTH_LONG).show();
      }
   };

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      binding = ActivityPractical22Binding.inflate(getLayoutInflater());
      setContentView(binding.getRoot());
      // Obtain the SupportMapFragment and get notified when the
map is ready to be used.
      SupportMapFragment mapFragment = (SupportMapFragment)
getSupportFragmentManager()
            .findFragmentById(R.id.map);
      mapFragment.getMapAsync(this);
      locationManager = (LocationManager)
getSystemService(Context.LOCATION_SERVICE);
   }

   @Override
   public void onMapReady(GoogleMap googleMap) {
      mMap = googleMap;
```

```
        addPointer();
    }
    private void addPointer() {
        if (ContextCompat.checkSelfPermission(this, android.Manifest.permission.ACCESS_FINE_LOCATION) != PackageManager.PERMISSION_GRANTED &&
                ContextCompat.checkSelfPermission(this, android.Manifest.permission.ACCESS_COARSE_LOCATION) != PackageManager.PERMISSION_GRANTED) {
            ActivityCompat.requestPermissions(this, new String[]{android.Manifest.permission.ACCESS_FINE_LOCATION, android.Manifest.permission.ACCESS_COARSE_LOCATION}, LOCATION_PERMISSION_REQUEST_CODE);
        } else {
            Location location = locationManager.getLastKnownLocation(LocationManager.GPS_PROVIDER);
            LatLng initialLocation = new LatLng(location.getLatitude(),location.getLongitude());
            currentMarker = mMap.addMarker(new MarkerOptions().position(initialLocation).title("Initial Location"));

mMap.moveCamera(CameraUpdateFactory.newLatLngZoom(initialLocation, 15));
locationManager.requestLocationUpdates(LocationManager.GPS_PROVIDER, 1, 1, locationListener);
        }
    }
    @Override
    public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions, @NonNull int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);
        if (requestCode == LOCATION_PERMISSION_REQUEST_CODE) {
            if (grantResults.length > 0 && grantResults[0] == PackageManager.PERMISSION_GRANTED) {
                // Obtain the SupportMapFragment and get notified when the map is ready to be used.
                SupportMapFragment mapFragment = (SupportMapFragment) getSupportFragmentManager()
                        .findFragmentById(R.id.map);
                mapFragment.getMapAsync(this);
            }else{
                Toast.makeText(this, "location permission required to add pointer", Toast.LENGTH_SHORT).show();
            }
        }
    }
}
```

Output:



Practical-23

AIM: Integrate Google maps API to your Android application and find the distance of any nearby location from your current location and display it.

Steps to follow:

1. Create a new project
2. Then right click on app > new > google > google maps activity
3. In Manifeast file you see a todo and in paste your api key in the api key section
4. You can use below api key if you don't have any
5. <meta-data
   android:name="com.google.android.geo.API_KEY"
   android:value="AIzaSyDV2_xy58r15K6TskZy4KWMuhUDVq67jqM" />

XML:

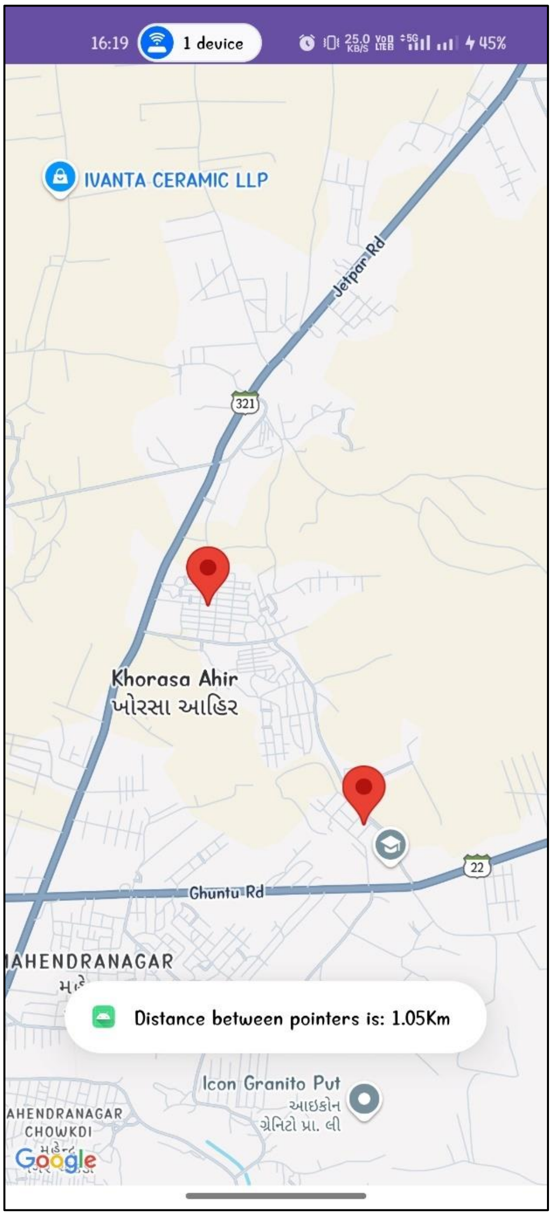
```
<?xml version="1.0" encoding="utf-8"?>
<fragment
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:map="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/map"

android:name="com.google.android.gms.maps.SupportMapFragme
nt"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".Practical23" />
```

JAVA:

```
package com.mk.madpractical;
import androidx.annotation.NonNull;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
import androidx.fragment.app.FragmentActivity;
import android.content.Context;
import android.content.pm.PackageManager;
import android.location.Location;
import android.location.LocationManager;
import android.os.Bundle;
import android.widget.Toast;
import com.google.android.gms.maps.CameraUpdateFactory;
import com.google.android.gms.maps.GoogleMap;
import com.google.android.gms.maps.OnMapReadyCallback;
import com.google.android.gms.maps.SupportMapFragment;
import com.google.android.gms.maps.model.LatLng;
import com.google.android.gms.maps.model.MarkerOptions;
import
com.mk.madpractical.databinding.ActivityPractical23Binding;
import com.google.android.gms.maps.model.Marker;

public class Practical23 extends FragmentActivity implements
OnMapReadyCallback {

   private GoogleMap mMap;
   private ActivityPractical23Binding binding;
   private final double Rad = 6371000;
   private LocationManager locationManager;
   private double lastLat,lastLng;
   private static final int LOCATION_PERMISSION_REQUEST_CODE =
1;
   private Marker newMarker;


   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);

      binding = ActivityPractical23Binding.inflate(getLayoutInflater());
      setContentView(binding.getRoot());
      locationManager = (LocationManager)
getSystemService(Context.LOCATION_SERVICE);

     getlocation();
   }
   @Override
   public void onMapReady(GoogleMap googleMap) {
      mMap = googleMap;
      LatLng initialLocation = new LatLng(lastLat,lastLng);
      mMap.addMarker(new
MarkerOptions().position(initialLocation).title("Your Last Location"));

mMap.moveCamera(CameraUpdateFactory.newLatLngZoom(initialL
ocation, 15));
      newMarker = mMap.addMarker(new
MarkerOptions().position(initialLocation).title("Your new Location"));
      mMap.setOnMapClickListener(latLng -> {
         newMarker.setPosition(latLng);

         double distance =
calculateDistance(latLng.latitude,latLng.longitude,lastLat,lastLng);
         Toast.makeText(this, "Distance between pointers is:
"+String.format("%.2f", distance/1000)+"Km",
Toast.LENGTH_SHORT).show();
      });
   }
   private double calculateDistance(double lat1, double lon1, double
lat2, double lon2) {
      // Convert latitude and longitude from degrees to radians
      double lat1Rad = Math.toRadians(lat1);
      double lat2Rad = Math.toRadians(lat2);
      double deltaLat = Math.toRadians(lat2 - lat1);
      double deltaLon = Math.toRadians(lon2 - lon1);

      // Haversine formula
      double a = Math.sin(deltaLat / 2) * Math.sin(deltaLat / 2) +
            Math.cos(lat1Rad) * Math.cos(lat2Rad) *
                  Math.sin(deltaLon / 2) * Math.sin(deltaLon / 2);
```

```
        double c = 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));

        // Distance in meters
        return Rad * c;
    }
    @Override
    public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions, @NonNull int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);
        if (requestCode == LOCATION_PERMISSION_REQUEST_CODE) {
            if (grantResults.length > 0 && grantResults[0] == PackageManager.PERMISSION_GRANTED) {
                getlocation();
            }else{
                Toast.makeText(this, "location permission required to add pointer", Toast.LENGTH_SHORT).show();
            }
        }
    }

    private void getlocation() {
        if (ContextCompat.checkSelfPermission(this, android.Manifest.permission.ACCESS_FINE_LOCATION) != PackageManager.PERMISSION_GRANTED &&
                ContextCompat.checkSelfPermission(this, android.Manifest.permission.ACCESS_COARSE_LOCATION) != PackageManager.PERMISSION_GRANTED) {
            ActivityCompat.requestPermissions(this, new String[]{android.Manifest.permission.ACCESS_FINE_LOCATION, android.Manifest.permission.ACCESS_COARSE_LOCATION}, LOCATION_PERMISSION_REQUEST_CODE);
        } else {
            Location location = locationManager.getLastKnownLocation(LocationManager.GPS_PROVIDER);
            lastLat = location.getLatitude();
            lastLng = location.getLongitude();
            // Obtain the SupportMapFragment and get notified when the map is ready to be used.
            SupportMapFragment mapFragment = (SupportMapFragment) getSupportFragmentManager()
                    .findFragmentById(R.id.map);
            mapFragment.getMapAsync(this);
        }
    }
}
```

Output:



Practical-24

AIM: Develop an application which performs Login using the Google account of the user.

Steps to follow:

1. Follow the steps from this video to connect with firebase for google login
2. https://www.youtube.com/watch?v=YQ0fJUiOYbY

XML:

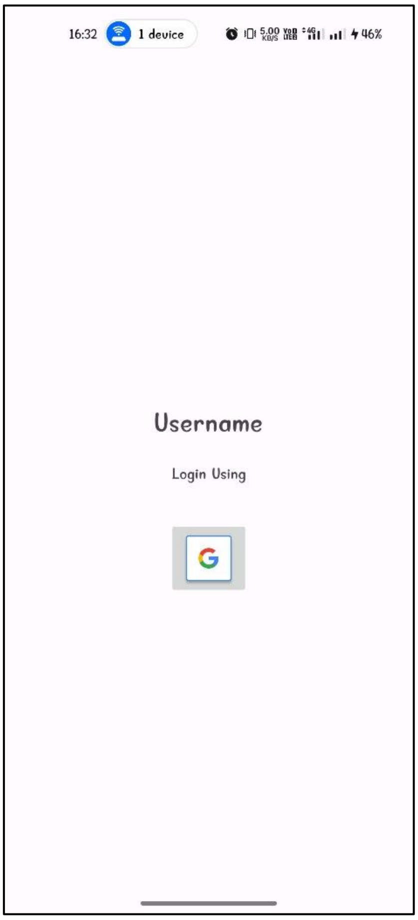
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/Practical24layout"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".Practical24">


   <TextView
      android:id="@+id/Practical24username"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_marginBottom="20dp"
      android:text="Username"
      android:textSize="24sp"

      app:layout_constraintBottom_toTopOf="@+id/Practical24txt"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintHorizontal_bias="0.498"
      app:layout_constraintStart_toStartOf="parent" />

   <TextView
      android:id="@+id/Practical24txt"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Login Using"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintHorizontal_bias="0.5"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent"
      app:layout_constraintVertical_bias="0.5" />

   <ImageButton
      android:id="@+id/Practical24login"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_marginTop="32dp"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintHorizontal_bias="0.5"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toBottomOf="@+id/Practical24txt"

app:srcCompat="@drawable/common_google_signin_btn_icon_dark
" />


</androidx.constraintlayout.widget.ConstraintLayout>
```

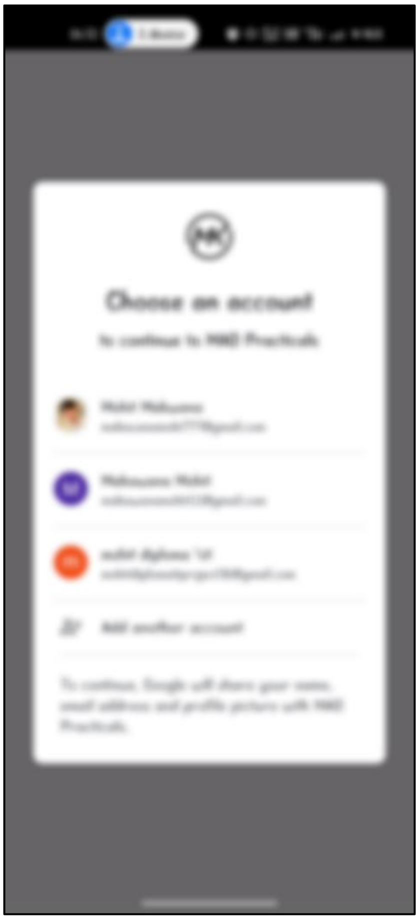
JAVA:

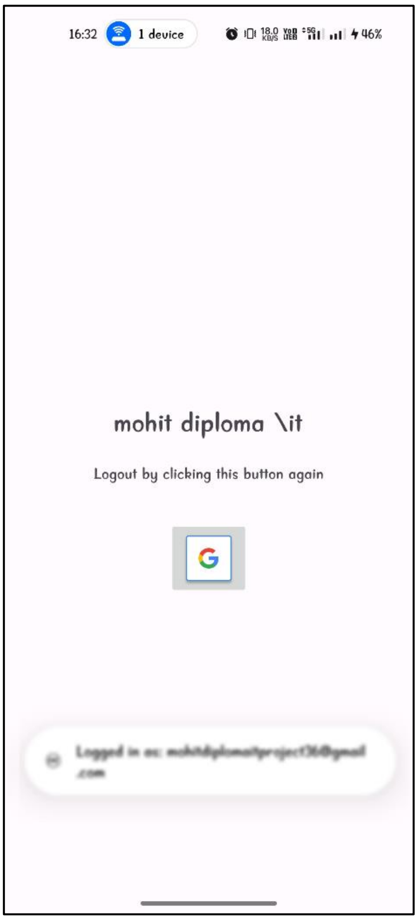
```
package com.mk.madpractical;
import android.content.Intent;
import android.os.Bundle;
import android.widget.ImageButton;
import android.widget.ImageView;
import android.widget.TextView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.activity.result.ActivityResultLauncher;
import androidx.activity.result.contract.ActivityResultContracts;
import androidx.annotation.Nullable;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import com.google.android.gms.auth.api.signin.GoogleSignIn;
import
com.google.android.gms.auth.api.signin.GoogleSignInAccount;
import com.google.android.gms.auth.api.signin.GoogleSignInClient;
import
com.google.android.gms.auth.api.signin.GoogleSignInOptions;
import com.google.android.gms.common.api.ApiException;
import com.google.firebase.auth.FirebaseAuth;
import com.google.firebase.auth.FirebaseUser;
import com.google.firebase.auth.GoogleAuthProvider;

public class Practical24 extends AppCompatActivity {

   private FirebaseAuth firebaseAuth;
   private GoogleSignInClient signInClient;
   private GoogleSignInOptions gso;
   private ImageButton btn;
   private TextView username,txt;
   FirebaseUser currentUser;

   // Register the launcher for Activity Result API
   private final ActivityResultLauncher<Intent> signInLauncher =
registerForActivityResult(
         new ActivityResultContracts.StartActivityForResult(),
         result -> {
            if (result.getResultCode() == RESULT_OK) {
               // Handle the Google Sign-In result
               Intent data = result.getData();
               handleSignInResult(data);
            } else {
               Toast.makeText(Practical24.this, "Sign-in failed",
Toast.LENGTH_SHORT).show();
            }
         }
   );

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
```

```
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical24);


ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Pra
ctical24layout), (v, insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
            return insets;
        });

        // Initialize Firebase Auth and Google Sign-In client
        firebaseAuth = FirebaseAuth.getInstance();
        gso = new
GoogleSignInOptions.Builder(GoogleSignInOptions.DEFAULT_SIGN_IN
)
                .requestIdToken(getString(R.string.default_web_client_id))
                .requestEmail()
                .build();
        signInClient = GoogleSignIn.getClient(this, gso);

        // Set up the login/logout button
        btn = findViewById(R.id.Practical24login);
        username = findViewById(R.id.Practical24username);
        txt = findViewById(R.id.Practical24txt);

        currentUser = firebaseAuth.getCurrentUser();

        // Check if user is logged in
        if (currentUser != null) {
            username.setText(currentUser.getDisplayName());
            txt.setText("Logout by clicking this button again");
            Toast.makeText(Practical24.this, "Logged in as: " +
currentUser.getEmail(), Toast.LENGTH_SHORT).show();

        }
        btn.setOnClickListener(v -> {
            currentUser = firebaseAuth.getCurrentUser();

            // Check if user is logged in
            if (currentUser != null) {
                // User is logged in, perform logout
                firebaseAuth.signOut();
                signInClient.signOut().addOnCompleteListener(task -> {
                    Toast.makeText(Practical24.this, "Logged out successfully",
Toast.LENGTH_SHORT).show();
                });
                username.setText("Username");
                txt.setText("Login Using");
            } else {
                // User is not logged in, perform Google sign-in
                Intent signInIntent = signInClient.getSignInIntent();
                signInLauncher.launch(signInIntent);  // Use the
ActivityResultLauncher to start sign-in
            }
        });
    }

    // Handle sign-in result
    private void handleSignInResult(@Nullable Intent data) {
        try {
            GoogleSignInAccount account =
GoogleSignIn.getSignedInAccountFromIntent(data).getResult(ApiExce
ption.class);
            if (account != null) {
                // Firebase authentication with Google
                firebaseAuthWithGoogle(account.getIdToken());
            }
        } catch (ApiException e) {
            Toast.makeText(this, "Sign in failed",
Toast.LENGTH_SHORT).show();
        }
    }

    private void firebaseAuthWithGoogle(String idToken) {

firebaseAuth.signInWithCredential(GoogleAuthProvider.getCredential
(idToken, null))
                .addOnCompleteListener(this, task -> {
                    if (task.isSuccessful()) {
                        // Sign-in success, show a toast with the user's email
(username)
                        currentUser = firebaseAuth.getCurrentUser();
                        if (currentUser != null) {
                            Toast.makeText(Practical24.this, "Logged in as: " +
currentUser.getEmail(), Toast.LENGTH_SHORT).show();
                            username.setText(currentUser.getDisplayName());
                            txt.setText("Logout by clicking this button again");
                        }
                    } else {
                        Toast.makeText(Practical24.this, "Authentication failed",
Toast.LENGTH_SHORT).show();
                    }
                });
    }
}
```

Output:


16:32
1 device
46%
Username
Login Using




16:32
1 device
46%
mohit diploma \it
Logout by clicking this button again